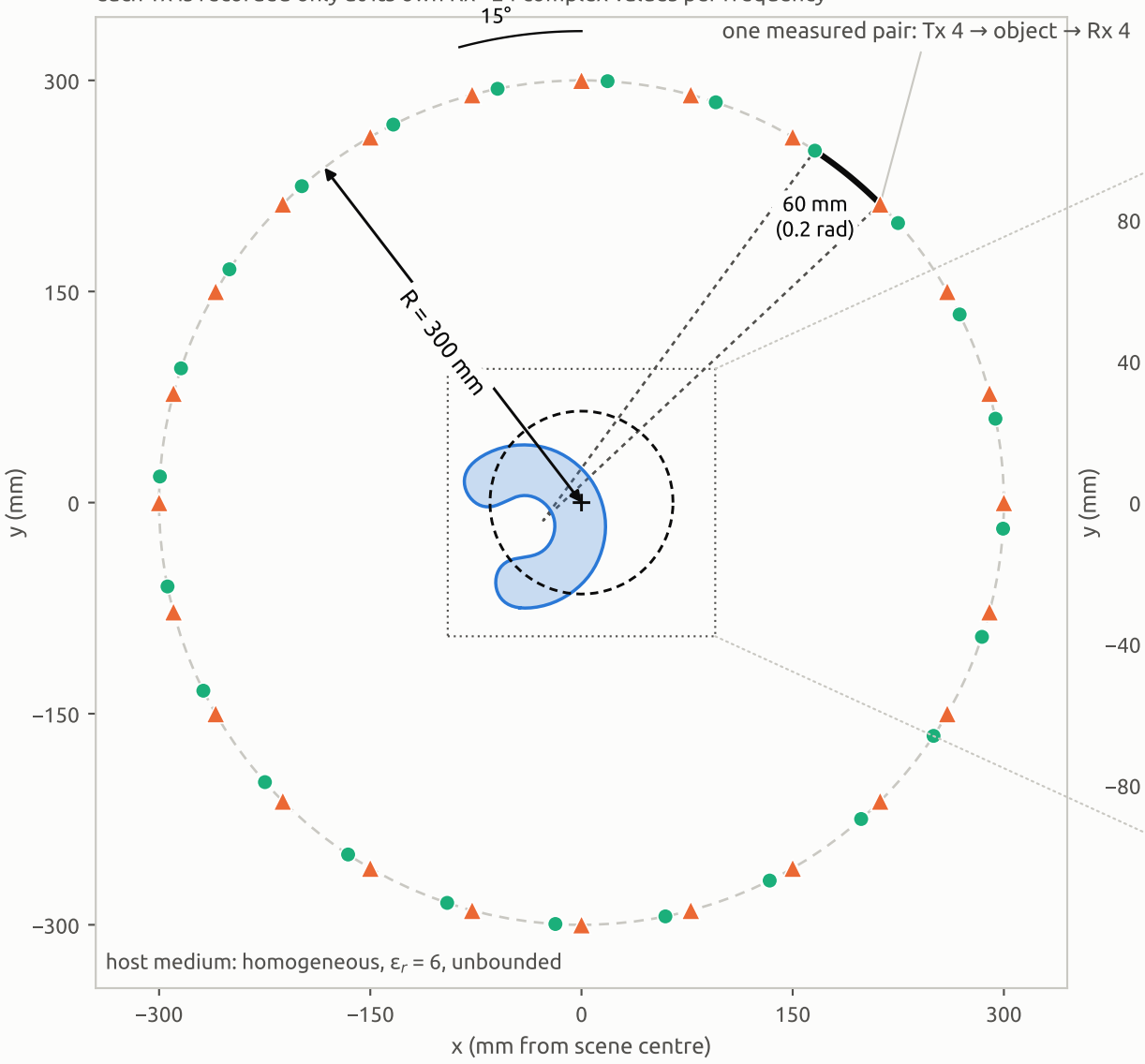
# TG-002 inverse-scattering setup

Scene shown: C-shape (same outline at contrasts 0.5 / 4 / 13.3). One start circle for all 30 cases. Drawn from the frozen input files; no new solves.

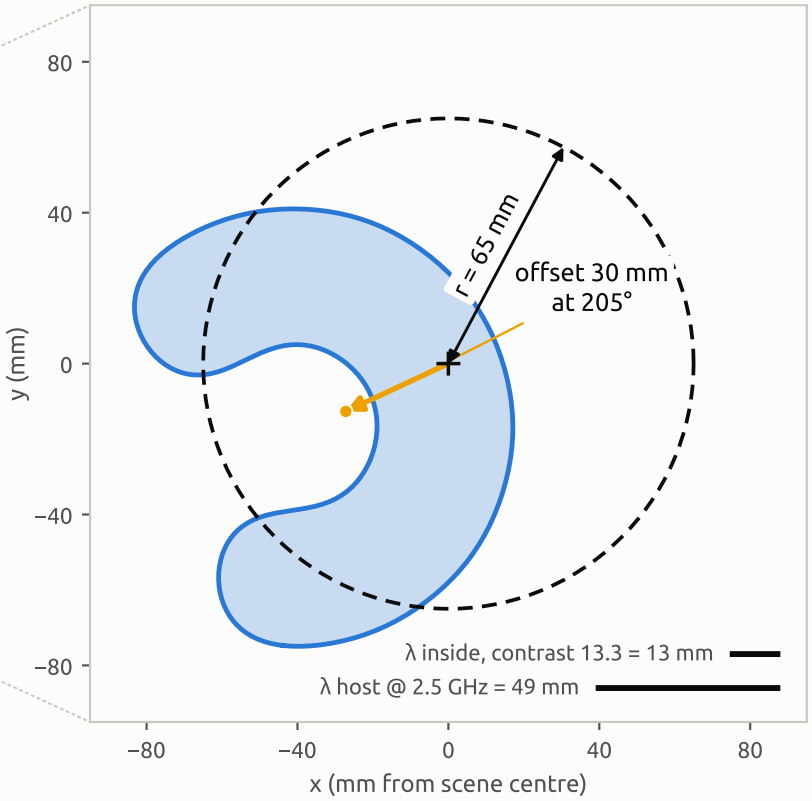
## Acquisition: 24 paired line sources and receivers

each Tx is recorded only at its own Rx · 24 complex values per frequency




## Unknown vs start: C-shape

rotation 0.3 rad · stored Fourier band K = 10




Tx (line source) · Rx (paired receiver) · true boundary · start circle · centroid offset · scene centre (0.5, 0.5) m

### Physics

2-D scalar Helmholtz transmission,
one homogeneous object,
known contrast $k_i^2/k_e^2$
0.5 / 4 / 13.3 (object $\varepsilon_r$ = 3 / 24 / 80)

### Frequencies

19 from 0.25 to 2.5 GHz, step 0.125
host λ = 490 → 49 mm
plus a damped copy k(1+0.25i)
for the low-frequency ladder

### Data

24 pairs × 19 freqs = 456 complex scattered-field values, noiseless; nodal Kress N2048, checked against N1024 to 1e-8

### Start and truth

start: circle r = 65 mm at centre
truths: 10 shapes, centroids 22–38 mm off centre

### Recovered if

RMS ≤ 1 mm, Hausdorff ≤ 2 mm, residual ≤ 0.3% at every freq., endpoint audit passed